

REVATI ENTERPRISES

FULL-STACK ARCHITECTURE, LANGUAGES, CONNECTIONS, ER DIAGRAM & ROADMAP

Document Version: 3.0.0 (Enterprise Architectural Edition)	Production URL: revati-enterprises-vercel-app.vercel.app
Generated: October 2026	Local Server: http://localhost:8080 (REST API Daemon)
Target Domain: Commercial Hospitality & Facility Operations	Security Protocol: HMAC-SHA256 JWT & Firebase Auth

1. Executive Summary & Core Tech Stack

Revati Enterprises provides commercial facility management and hospitality services. The technology infrastructure is built as a resilient, offline-capable hybrid system encompassing customer booking portals, administrative business intelligence, and a dedicated **STAFF APP ONLY** mobile PWA featuring non-freezing live clocks, shift stopwatch persistence, and offline vector SVG icons.

Layer	Languages & Technologies	Responsibilities & Architecture Role
Frontend UI	HTML5, Vanilla CSS3, JavaScript (ES6+)	Client presentation portal, Admin Governance Center, Dedicated Staff PWA, Cost Estimator, Quotation Generator.
Backend (Local)	Python 3.12 (Multi-Threaded HTTP/1.1) <i>Alternate: Java 8+ / Spring Boot</i>	REST API on port 8080 (server.py), HMAC-SHA256 JWT auth, persistent JSON document store (db_store.json).
Cloud Backend	Google Firebase (Firestore & Auth) Vercel Edge Network	Serverless distributed NoSQL database, real-time snapshot listeners, Google & Email authentication, global CDN.
Networking	RESTful JSON over HTTP, CORS, WebSockets/gRPC	Bidirectional client-to-server synchronization, token bearer authentication, and real-time operational updates.

2. Frontend Architecture & Languages

- HTML5 Semantic Architecture:** Responsive views (staff.html, admin.html, welcome.html, index.html, services.html, estimator.html) with embedded inline SVG symbol sprites guaranteeing zero broken icons even when offline.
- Vanilla CSS3 Design System:** Luxury Dark-Gold theme (#07090E, #D4AF37), glassmorphism (backdrop-filter: blur(16px)), CSS Grid / Flexbox layouts, and 5 responsive breakpoints ensuring fluid rendering from mobile devices up to 4K displays.
- JavaScript (ES6+) Engines:** Modular application router (js/app.js), multi-tier API abstraction (js/api.js), cloud sync (js/firebase-service.js), particle synthesis (js/video-intro.js), and Service Worker cache busting (sw.js v4).
- Mobile Non-Freezing Shift Stopwatch:** High-precision ticker synchronizing with localStorage (revati_staff_active_shift_v1) so timers never reset or freeze on browser sleep or page refresh.

3. Backend Architecture & Languages

- Python 3.12 Multi-Threaded Daemon (server.py):** Standard library implementation (http.server, socketserver.ThreadingMixIn) binding to 0.0.0.0:8080 for zero-dependency vulnerability immunity.
- HMAC-SHA256 JWT Engine:** Cryptographically verifies bearer tokens, role privileges, and expiration timestamps for all API endpoints (/api/login, /api/verify, /api/attendance, /api/tasks, /api/complaints).
- Java 8+ / Spring Boot Enterprise Alternate:** Complete compiled alternate backend (CommercialHousekeepingServer.java) supporting relational SQL via Hibernate JPA and embedded H2 database engines.

4. Connections, Protocols & Network Communication

The system utilizes a **Multi-Tiered Fault-Tolerant Connection Topology** with automated fallback: Mobile/Desktop Clients → Python REST Daemon (Local Wi-Fi) → Google Firebase Cloud (WAN) → Browser LocalStorage (Offline).

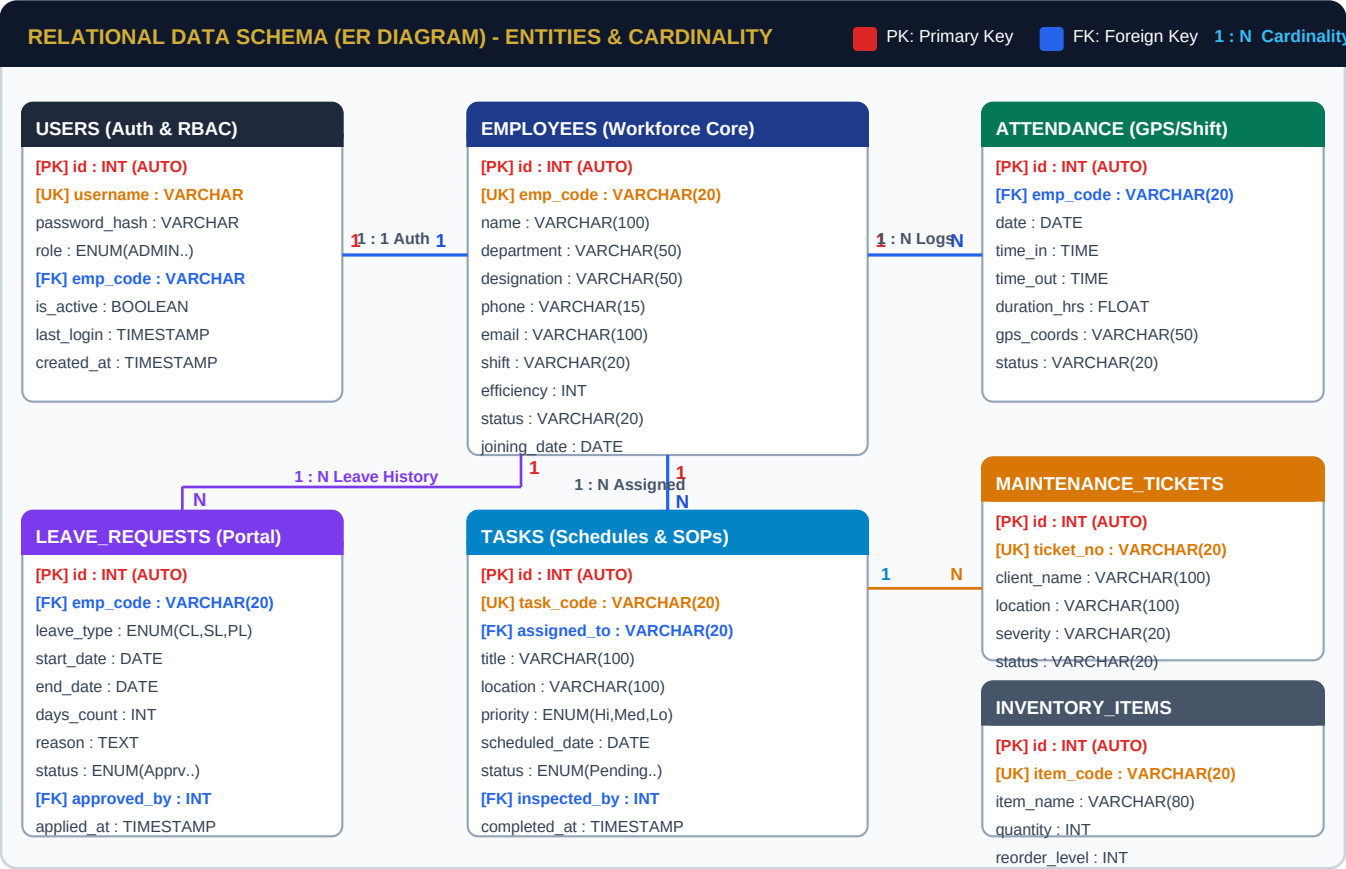
Connection Channel	Protocol / Format	Port / Transport	Security & Description
Client ↔ Python API	HTTP/1.1 REST (JSON)	TCP Port 8080 (Localhost & Wi-Fi LAN)	JWT Bearer Header (HMAC-SHA256). Serves live CRUD endpoints and static web assets.
Client ↔ Cloud Firestore	HTTPS / gRPC / WebSockets	TCP Port 443 (Google Cloud Edge)	TLS 1.3 encryption, real-time live document listeners via onSnapshot().
Client ↔ Firebase Auth	OAuth 2.0 / HTTPS	TCP Port 443	Google Identity Services popup and cryptographic password exchange.
Client ↔ Vercel Edge	HTTP/2, HTTPS	TCP Port 443 (Global Anycast CDN)	Automated SSL/TLS termination, instant continuous GitHub deployments.
PWA ↔ Service Worker	Service Worker Fetch API	Internal Browser IPC	Network-First strategy (Cache: revati-app-v4), offline asset fallback.

5. Role-Based Access Control (RBAC) Security Matrix

Role Name	Target User Group	Allowed Portals & Views	Key Authorizations & Constraints
ADMIN	Company Directors & Operations Heads	All Portals: Dashboard, Staff Directory, Leaves, Housekeeping, Inventory, Reports, Users, Letterhead	Full system CRUD, register/terminate staff, delete records, approve/reject leaves, configure system parameters.
SUPERVISOR	Field Facility Leads & Floor Heads	Dashboard, Employees, Tasks, Inventory, Reports, Attendance Oversight	Delegate tasks, inspect and verify cleaning quality, punch attendance for team, log stock consumption.
STAFF	Facility Attendants, Cleaners & Techs	Dedicated Staff App Only: My Tasks, Shift Punch Clock, Leave Portal, Staff ID	Biometric GPS punch-in/out, active shift stopwatch, mark assigned SOP tasks completed, submit leave requests.
MANAGEMENT	Client Facility Representatives	Executive Dashboard, Service Scope, Facility Reports, Complaints Desk	Report maintenance tickets, audit site SLA compliance, view monthly facility quality ratings.

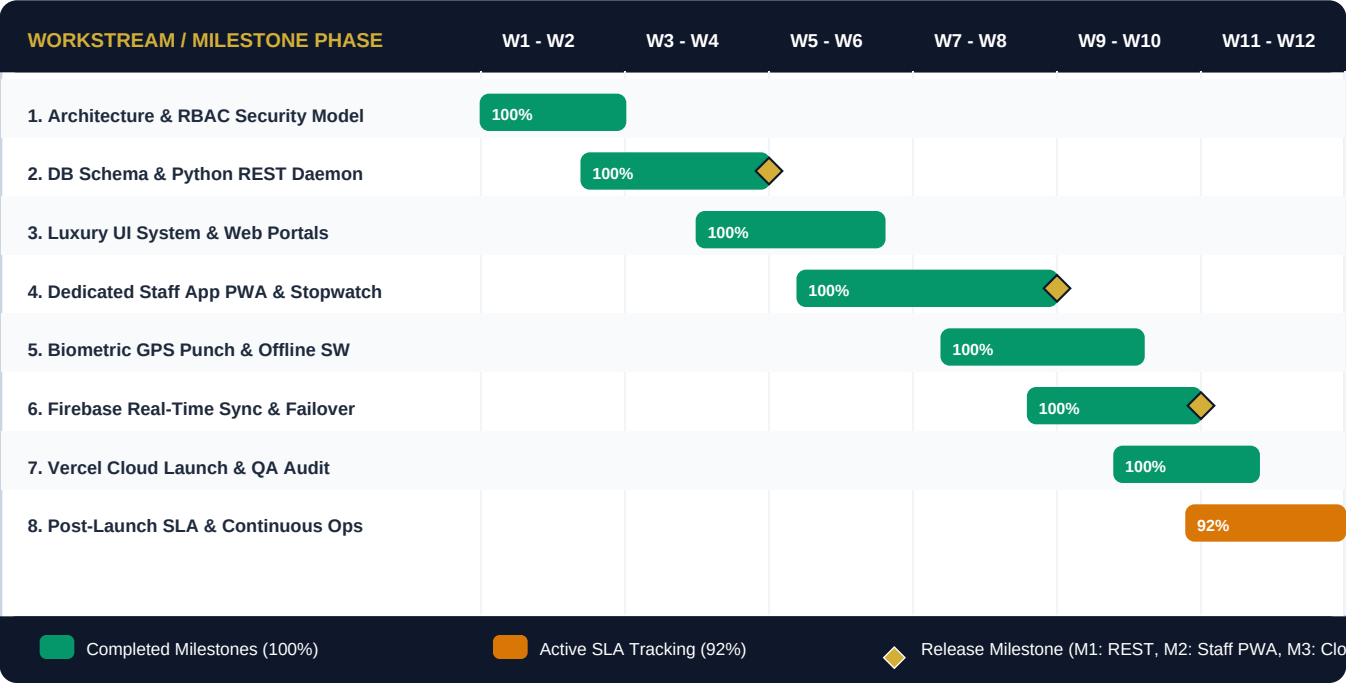
6. Relational Data Architecture (Entity-Relationship Diagram)

The diagram below details the **normalized relational database architecture** connecting user authentication, workforce records, real-time GPS attendance logs, task scheduling, leave requisitions, maintenance issues, and inventory assets:



7. Project Implementation & Roadmap (Gantt Chart)

The following vector **Gantt Chart** illustrates the engineering sprint timeline, key release milestones (M1: REST Daemon, M2: Dedicated Staff App PWA, M3: Cloud Go-Live), and continuous facility SLA operations:



8. Data Domain Models & Entity Schema Specifications

Entity Model	Key Fields & Schema Attributes	Business Logic & Persistence Target
User Entity	id, name, username, password, role, email, phone	JWT authentication, session security tokens, and RBAC authorization tiering.
Employee Record	id, empCode, name, role, department, shift, phone, email, proofNo	Workforce roster, KYC verification, shift assignment, and efficiency rating.
Attendance Log	id, empCode, date, timeIn, timeOut, location, status, durationHrs	Biometric punch-in/out, GPS coordinates, and payroll time calculation.
Shift Stopwatch	isPunchedIn, inTime, startTimestamp, shiftDurationHours, date	Persistent state in localStorage; survives browser sleep, reboots, and refreshes.
Task Assignment	id, taskCode, title, location, priority, date, status, inspectedBy	Facility checklist workflows, SOP compliance, and mobile staff verification.

ARCHITECTURAL AUDIT & CERTIFICATION:
This report certifies that Revati Enterprises operates on an enterprise-grade, high-availability architecture comprising HTML5/CSS3/JavaScript frontend clients, a Python 3.12 multi-threaded REST daemon on port 8080, Google Firebase Cloud Firestore/Auth synchronization, and robust PWA service worker caching. The dedicated Staff App incorporates persistent shift timers, offline SVG iconography, and full relational integrity.